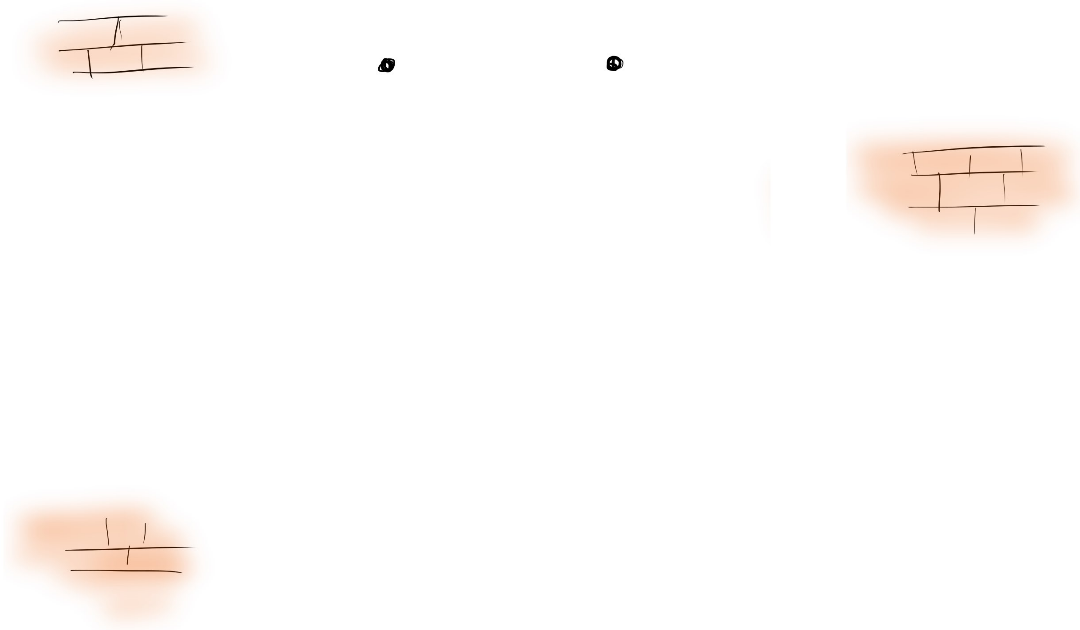
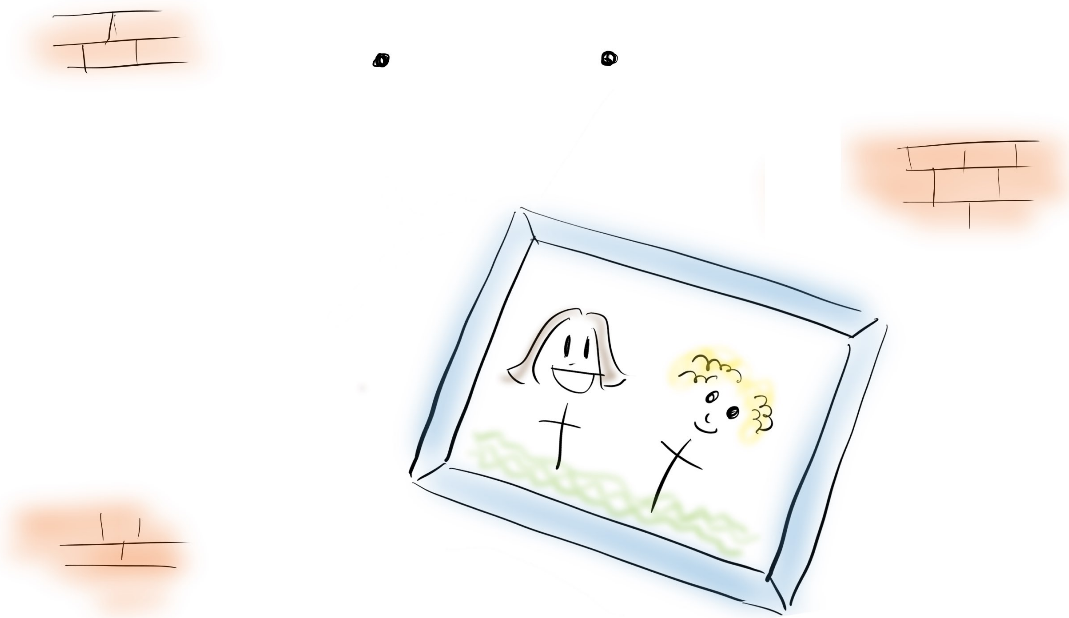
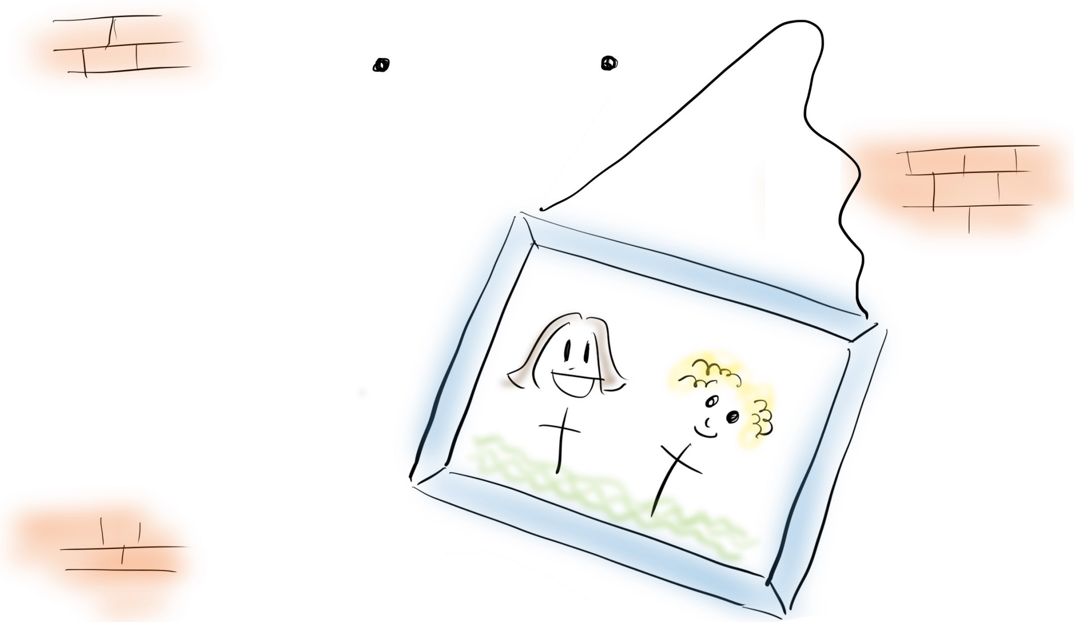
PICTURE-HANGING PUZZLES

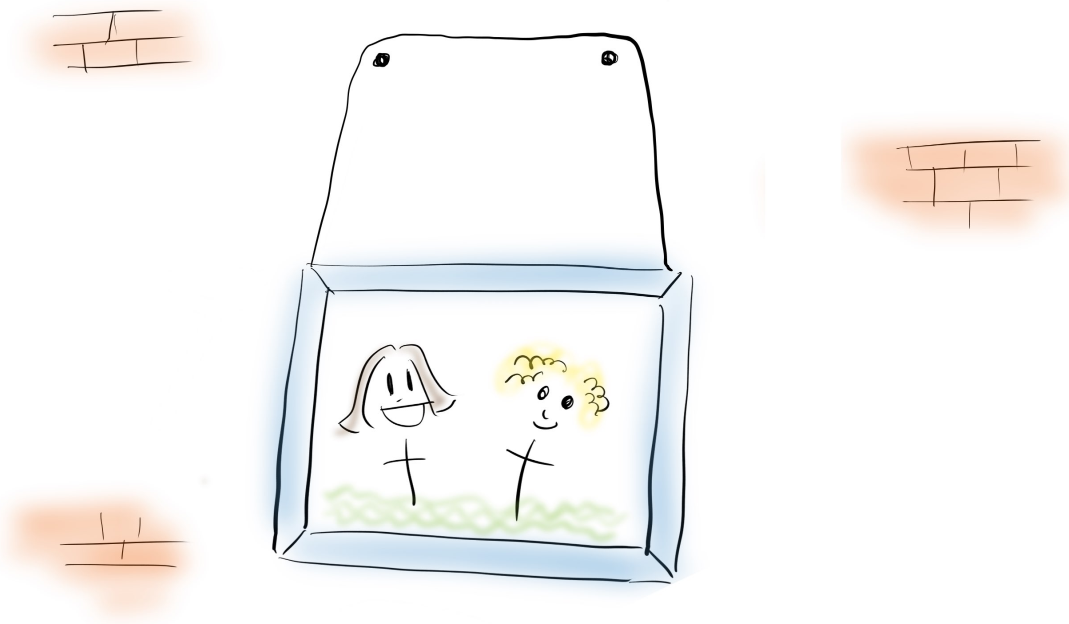
Lucía Asencio Martín
Newcastle University

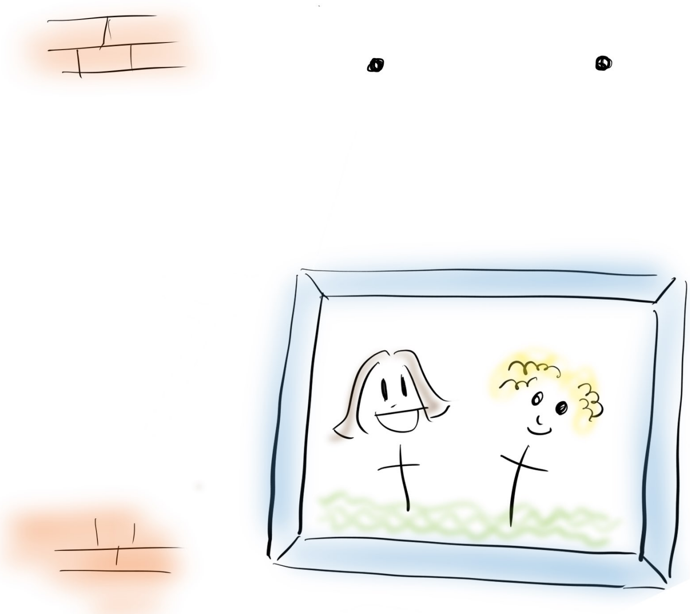




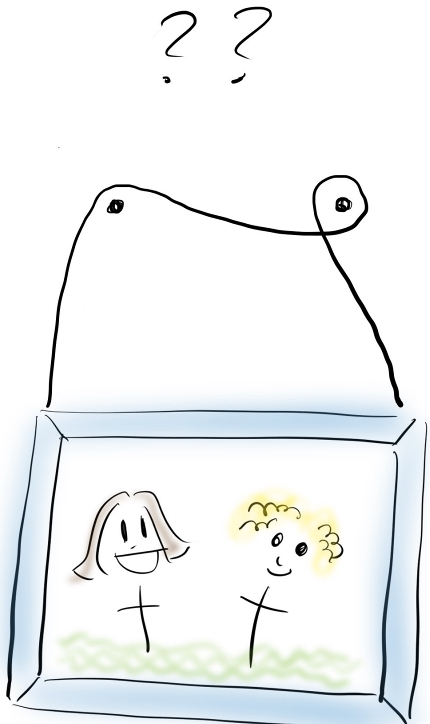


Normal people :





Hang picture so that it falls if you remove <u>any</u> of the 2 nails



Hang picture so that it falls if you remove *any* of the 2 nails

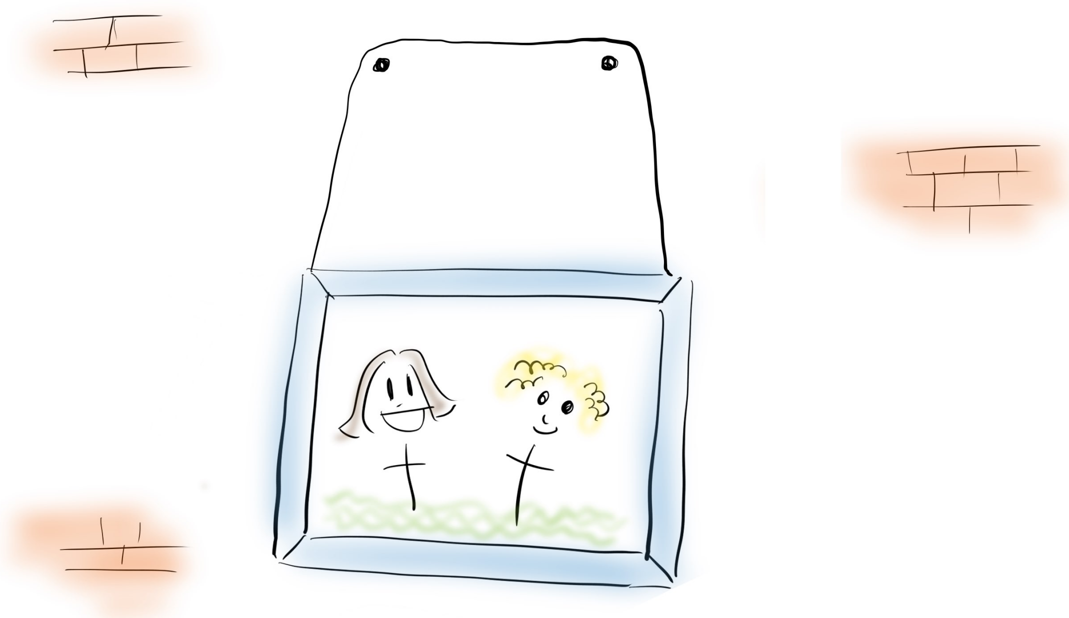
Hang picture

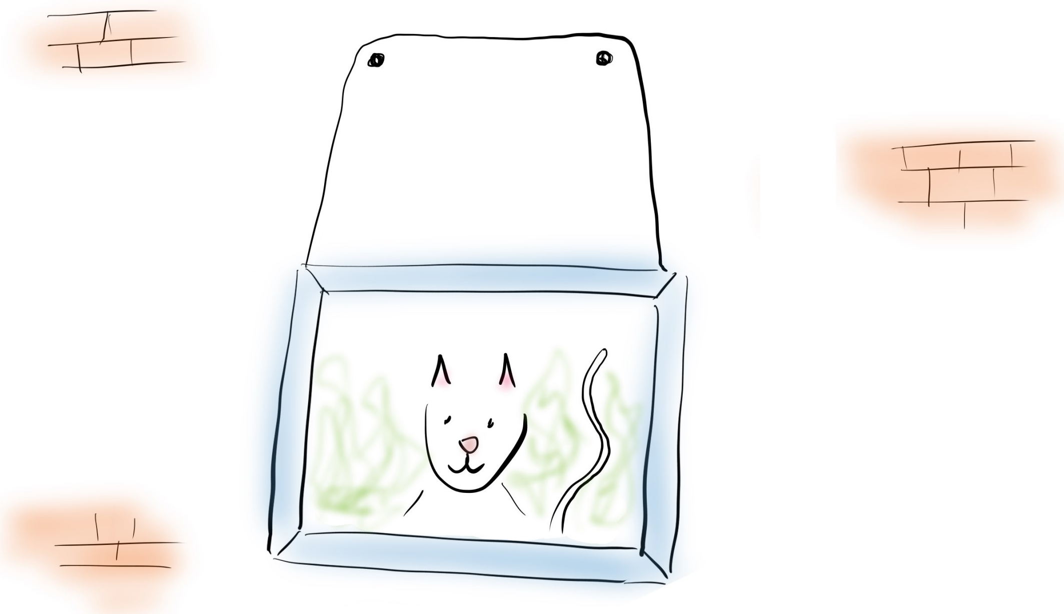
so that it falls

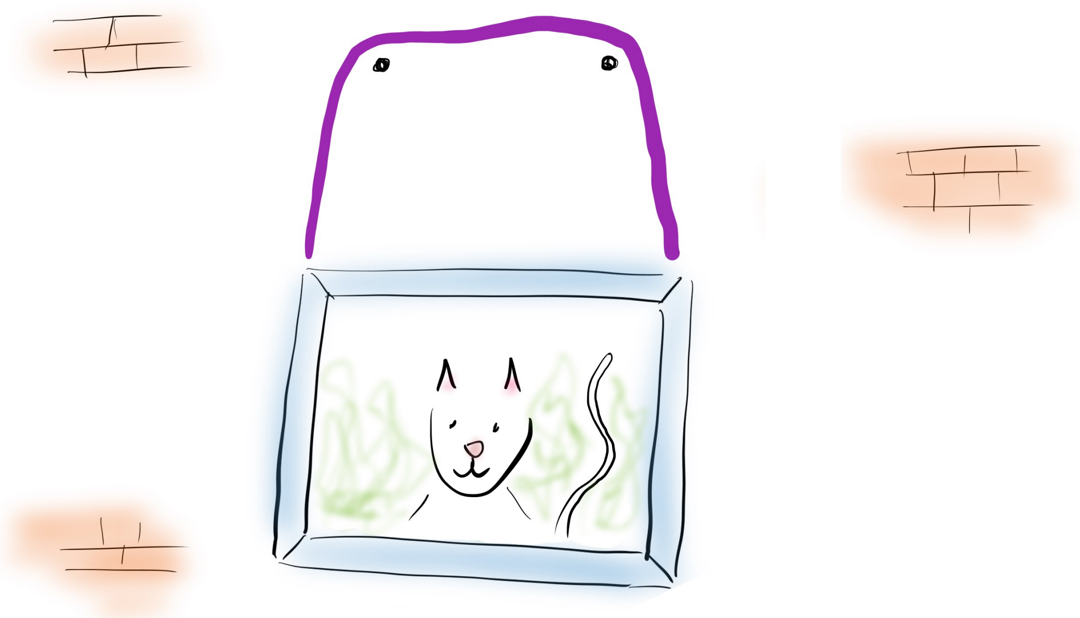
if you remove

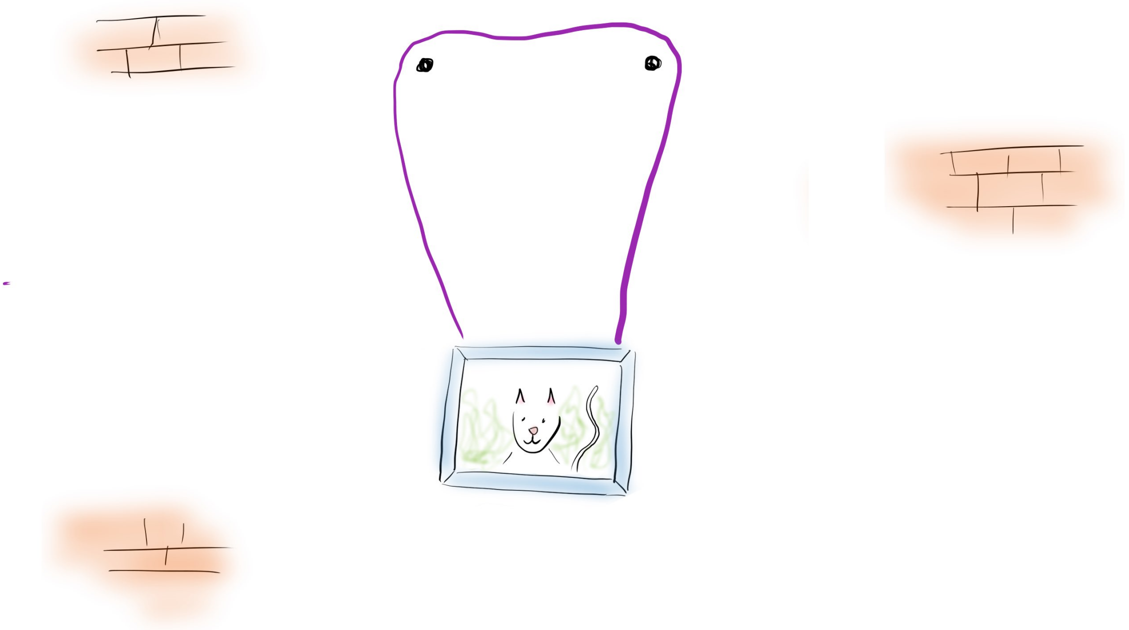
<u>any</u> of the 2 nails

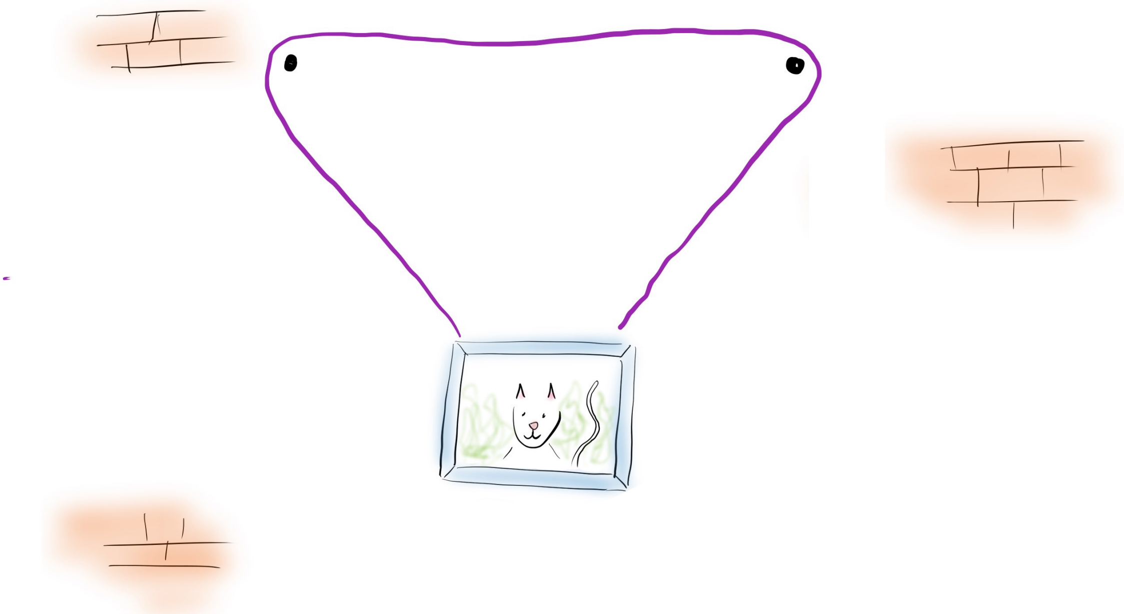
Can you do it ?



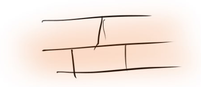


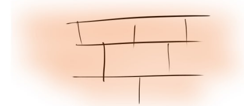


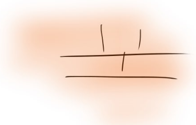


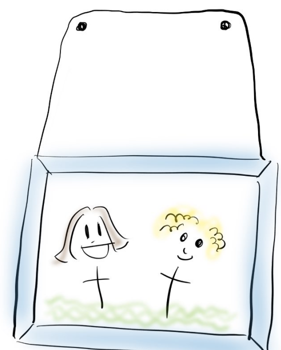


What is important?









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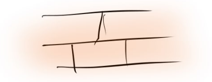
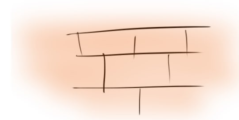
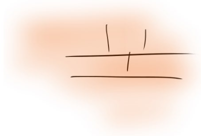
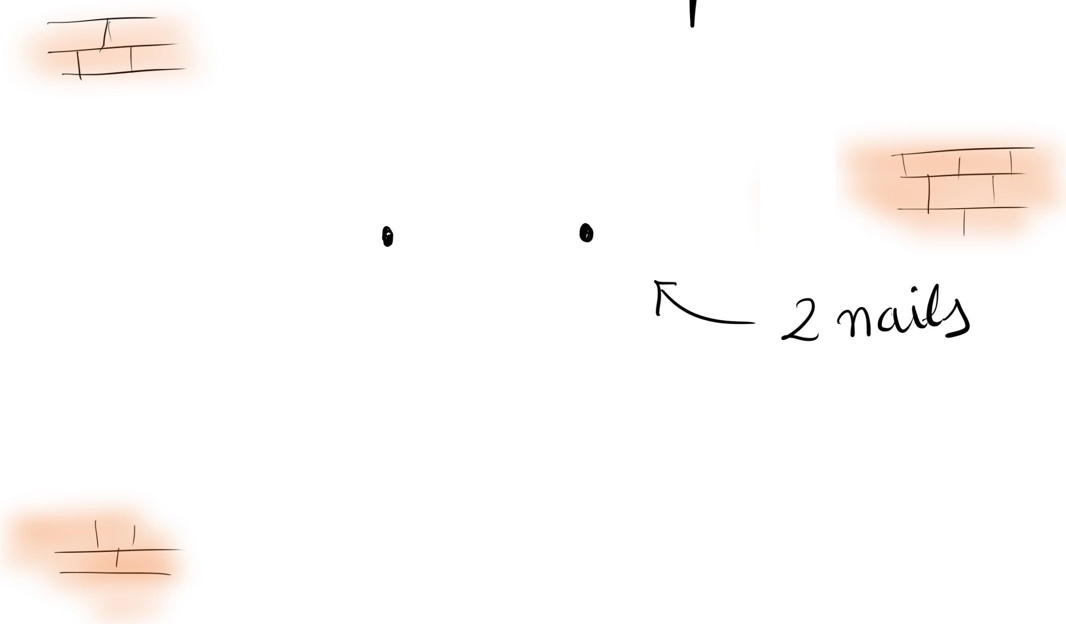


A wall





What is important?



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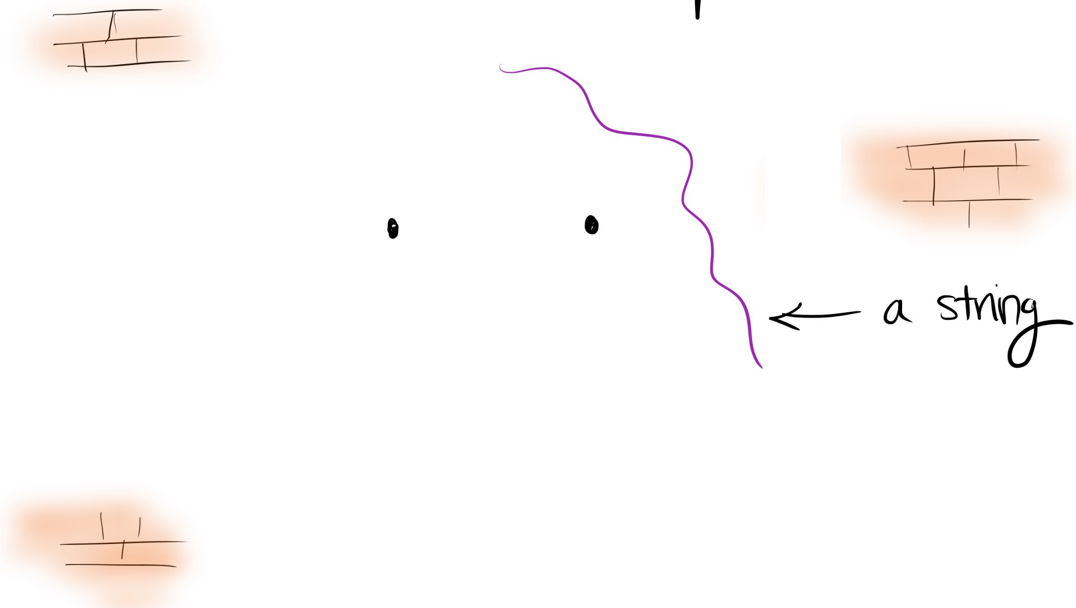

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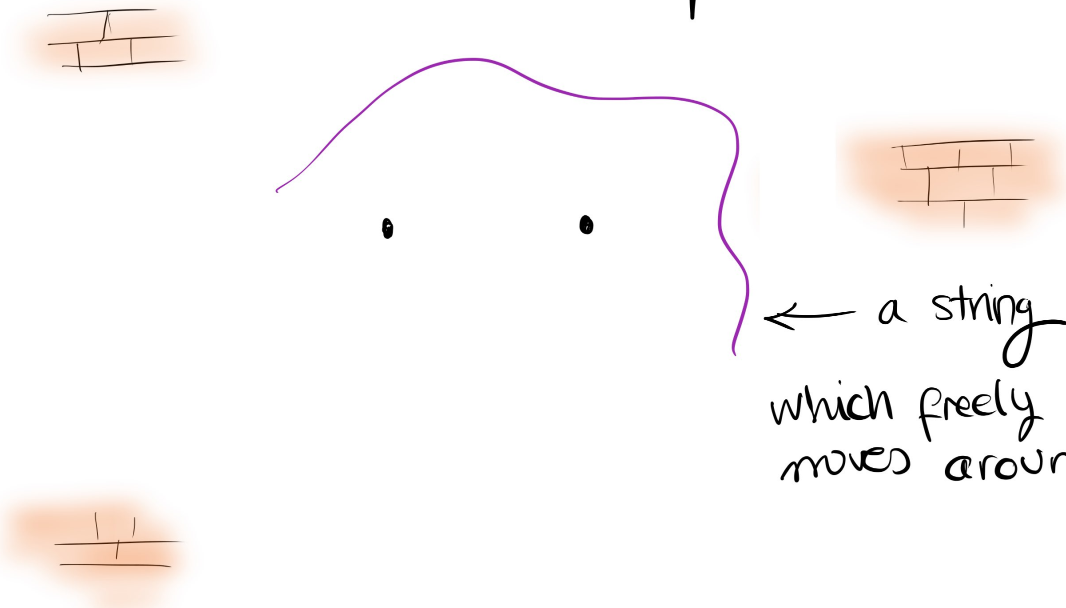

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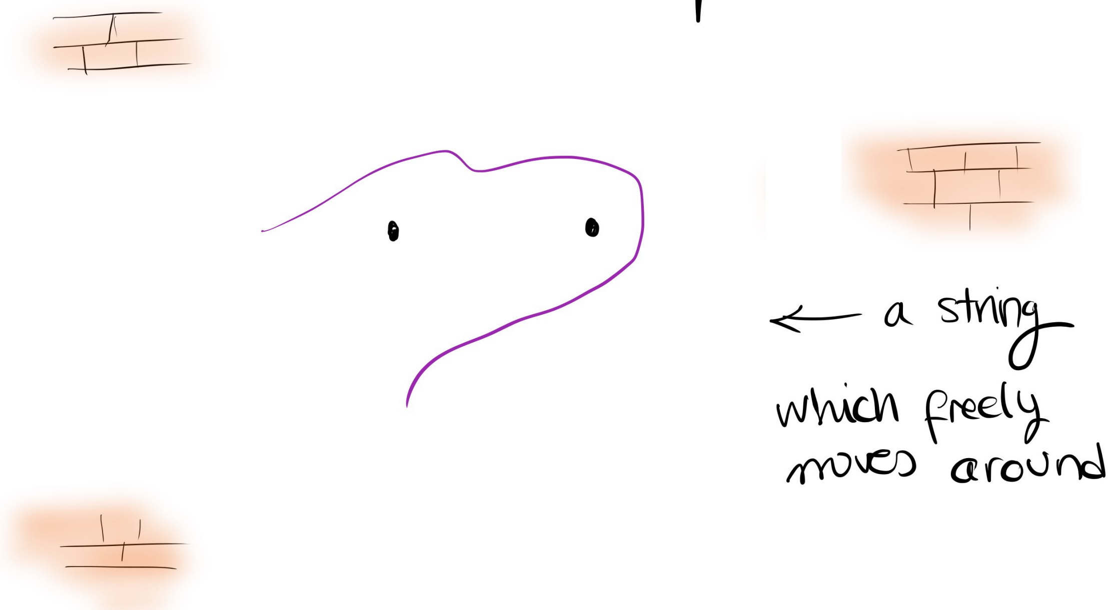

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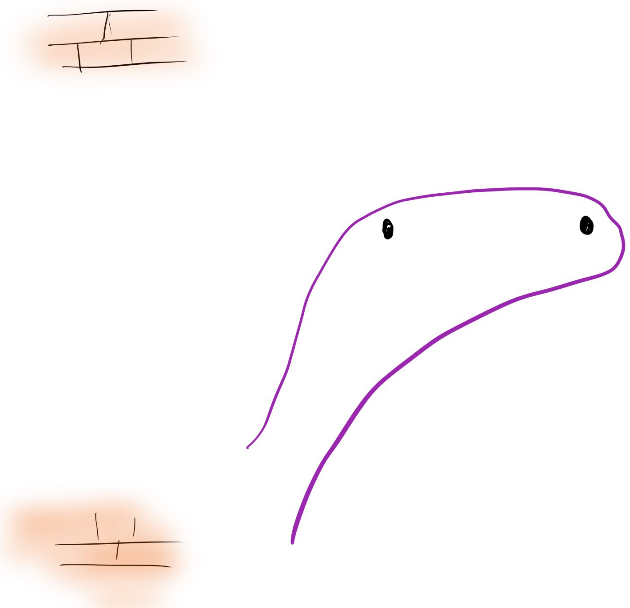
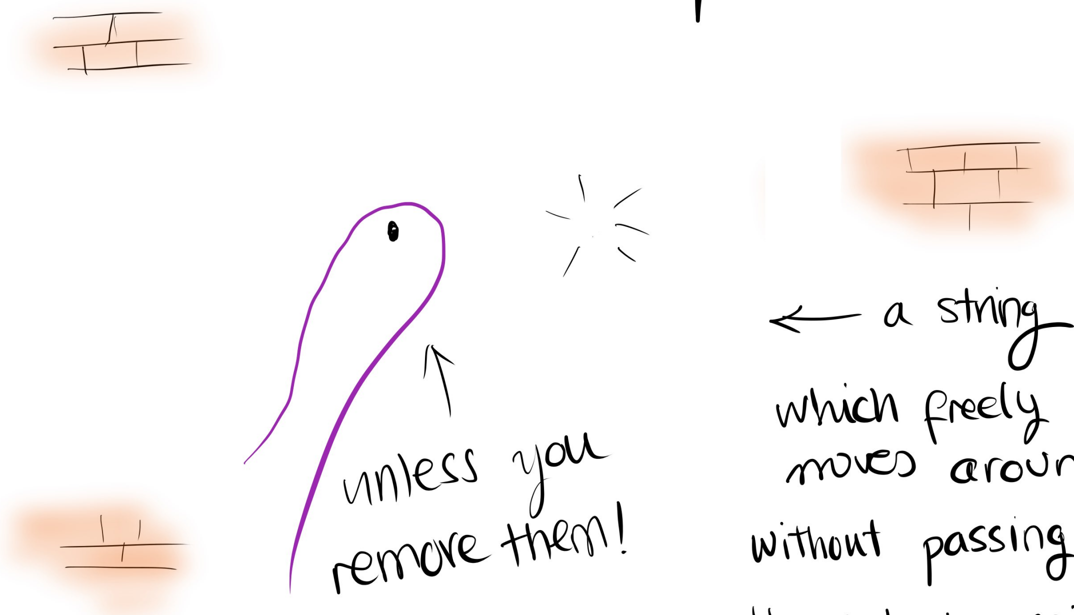


← a string which freely moves around without passing through the nails

What is important ?



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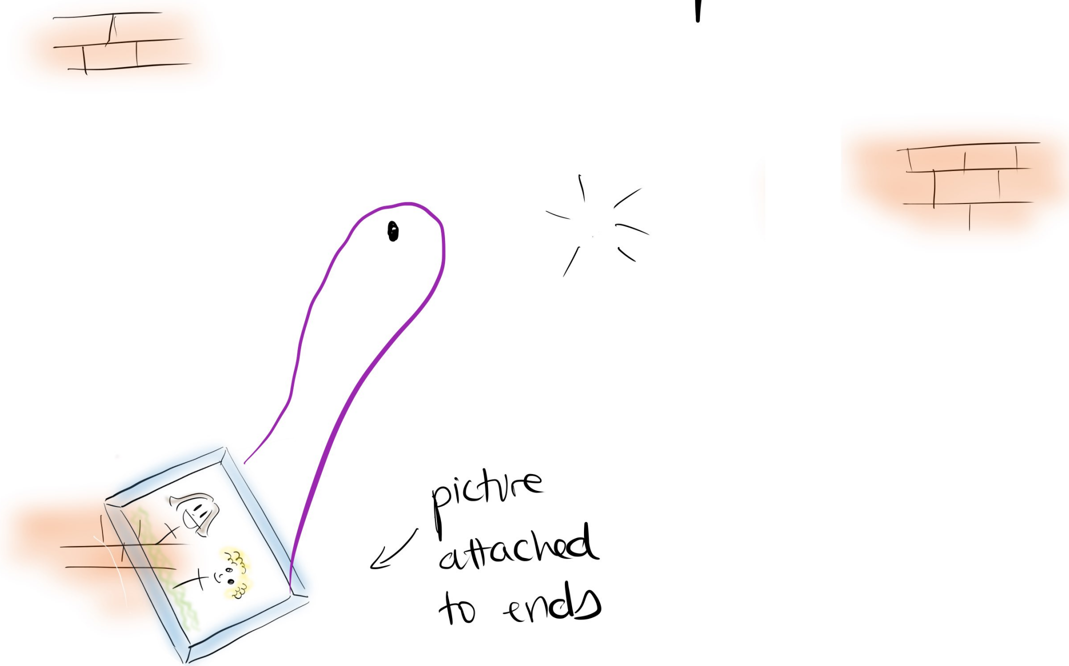
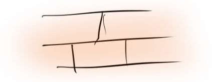
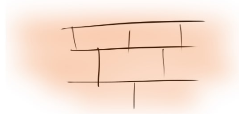
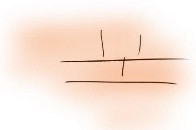


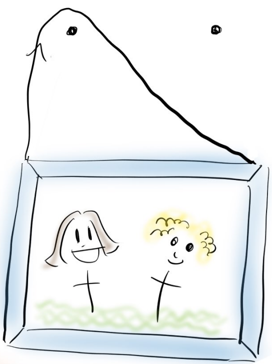
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



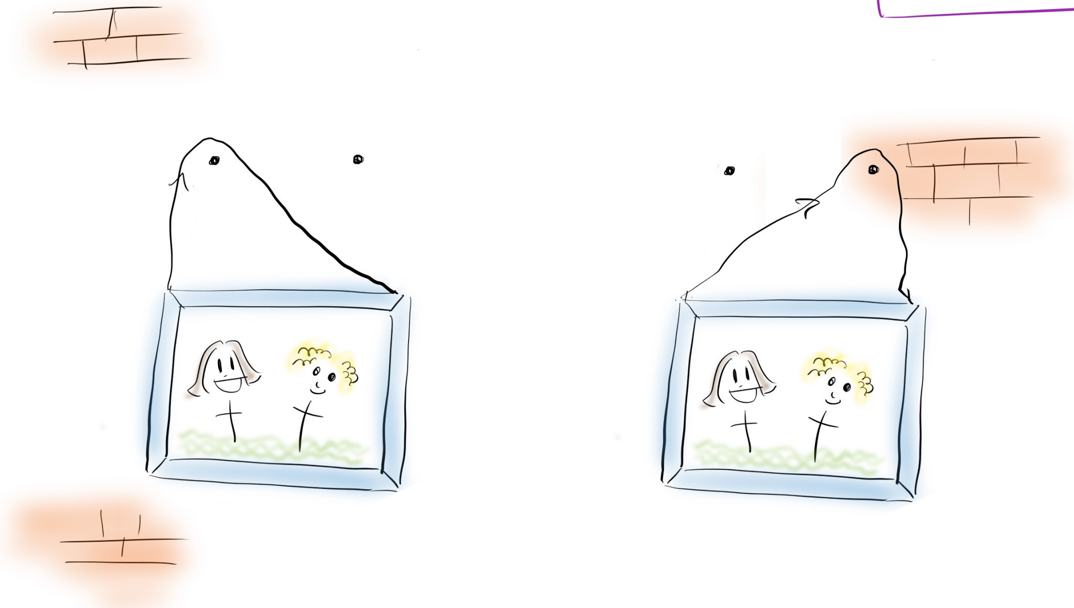




GOAL: Hang picture so that it falls if you remove <ins>any</ins> of the 2 nails

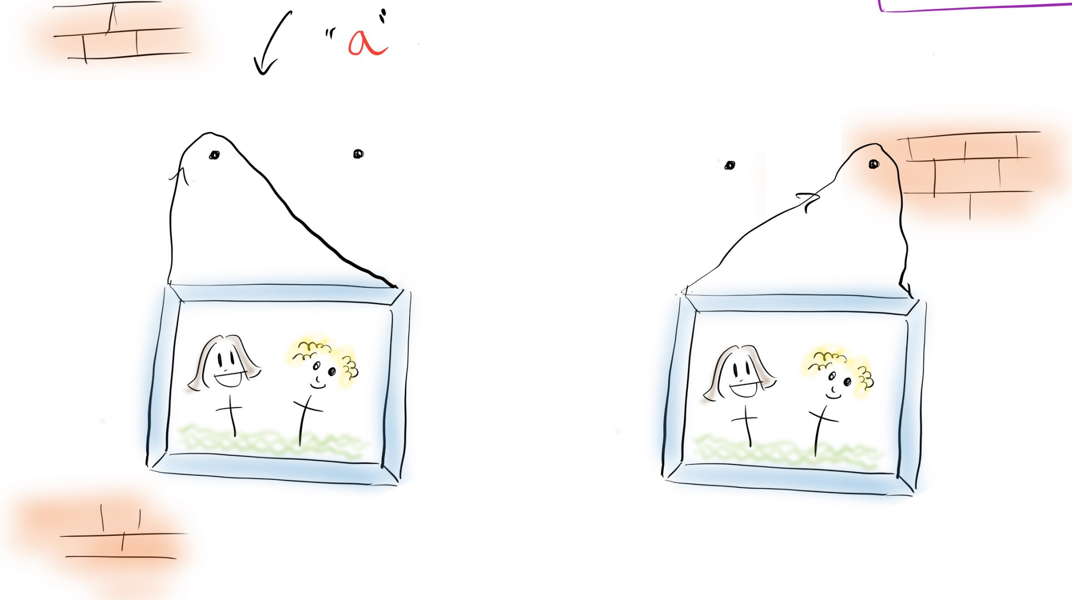


GOAL: Hang picture so that it falls if you remove any of the 2 nails

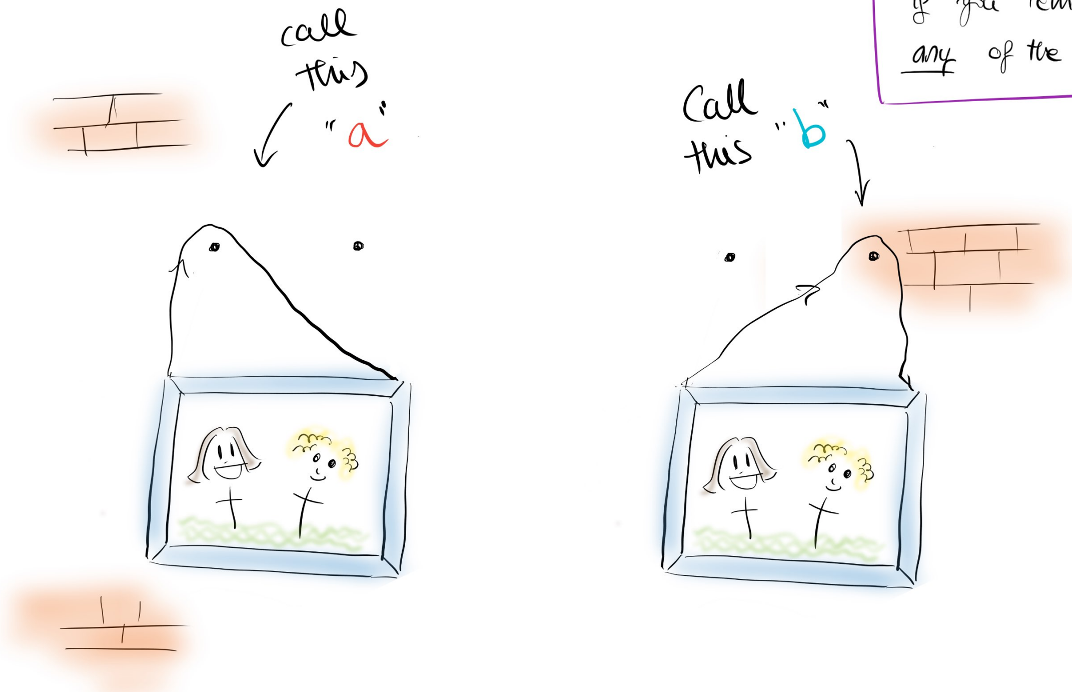


GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

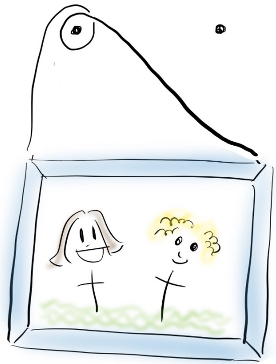
call this "a"



GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

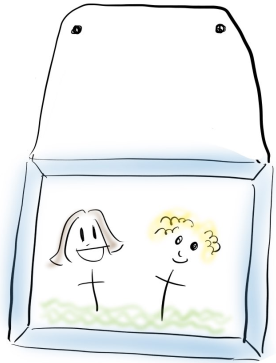


GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



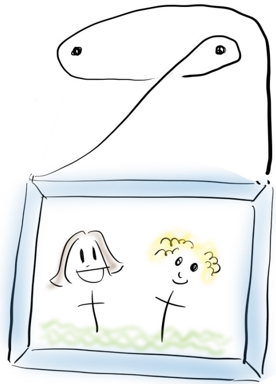
picture-hanging aa

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



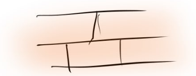
picture-hanging ab

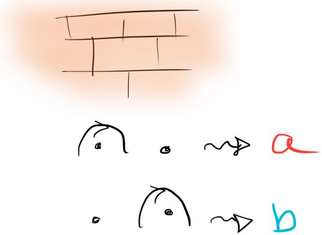
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

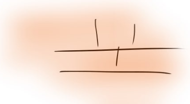


picture-hanging bab

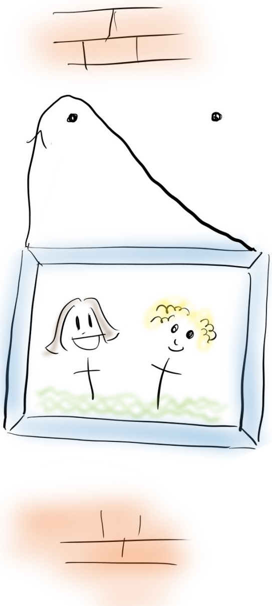
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







⚠ This ...



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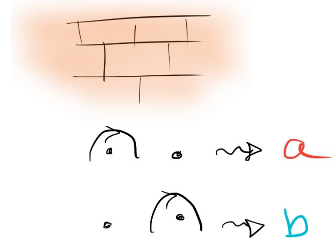
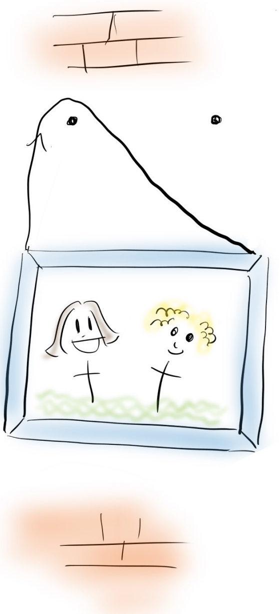
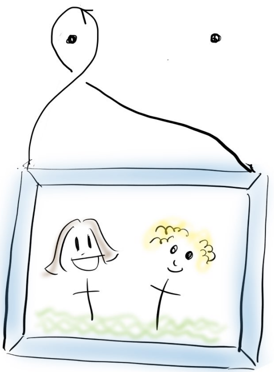
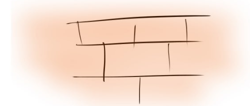


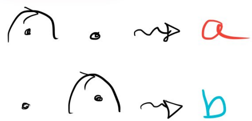
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

⚠ This ... different from this!









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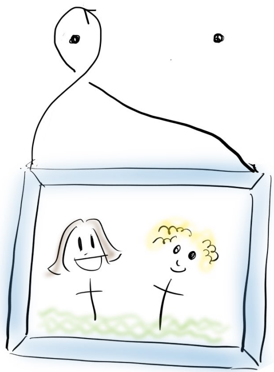
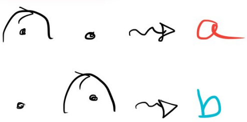
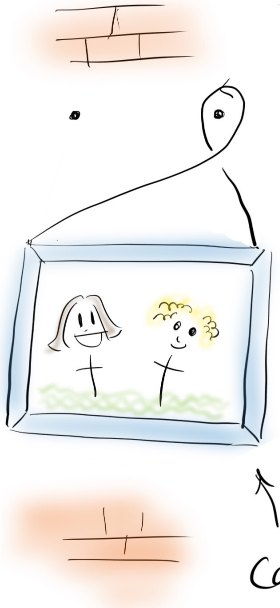
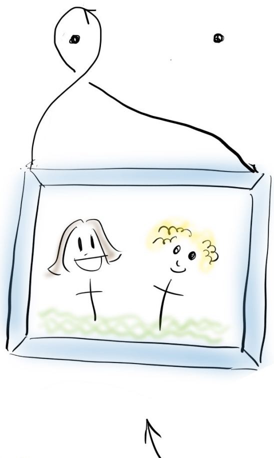


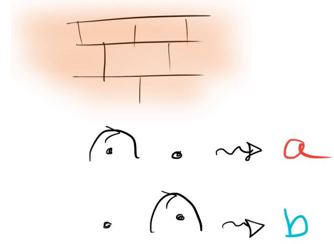


call this "a^{-1}"

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

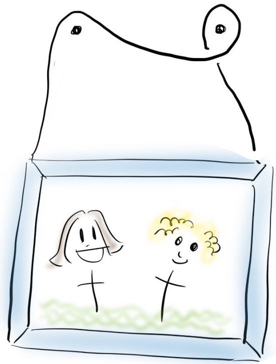




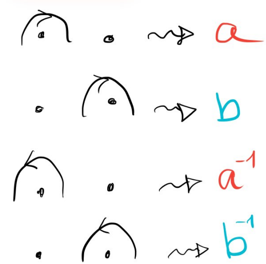


Call this "b^{-1}" call this "a^{-1}"

GOAL: Hang picture so that it falls if you remove *any* of the 2 nails

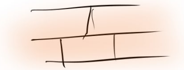


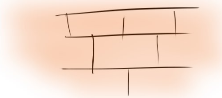
picture-hanging ab^{-1}

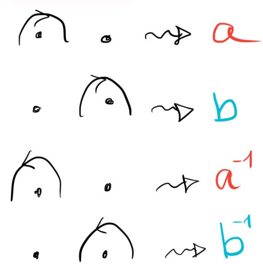


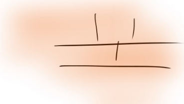
A falling picture-hanging?

GOAL: Hang picture so that it falls if you remove **any** of the 2 nails









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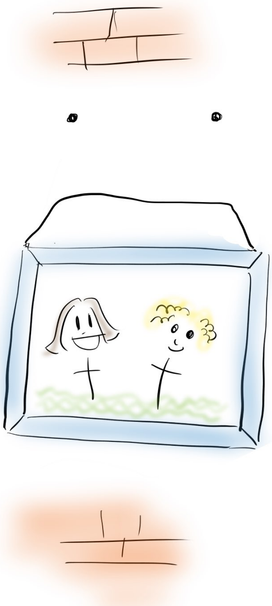

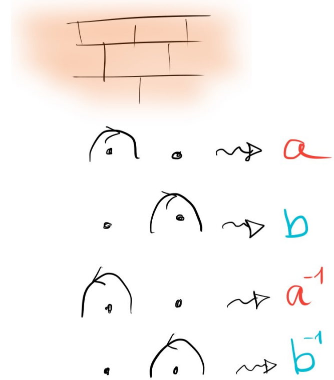

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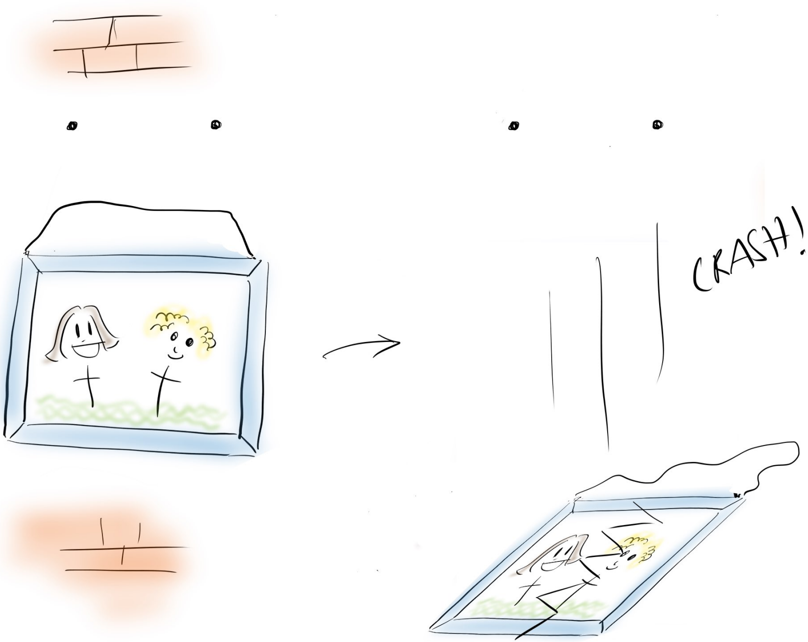

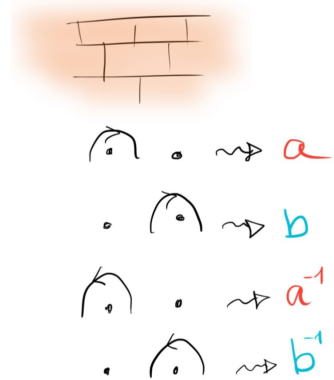

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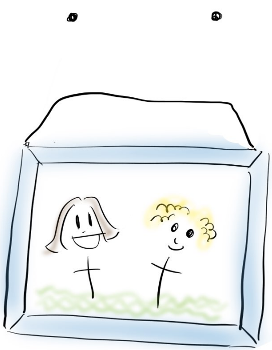
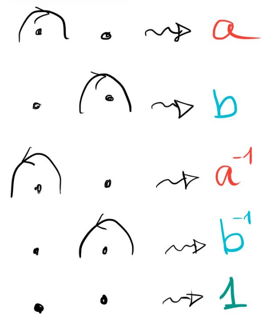


call this "1"



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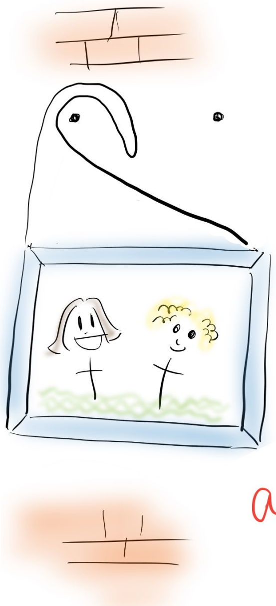
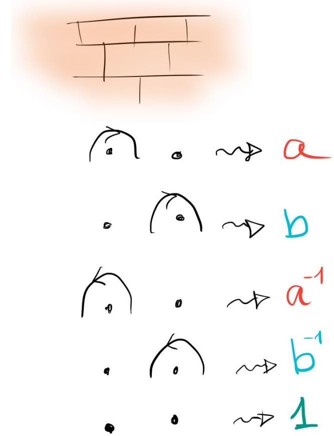


aa^{-1}



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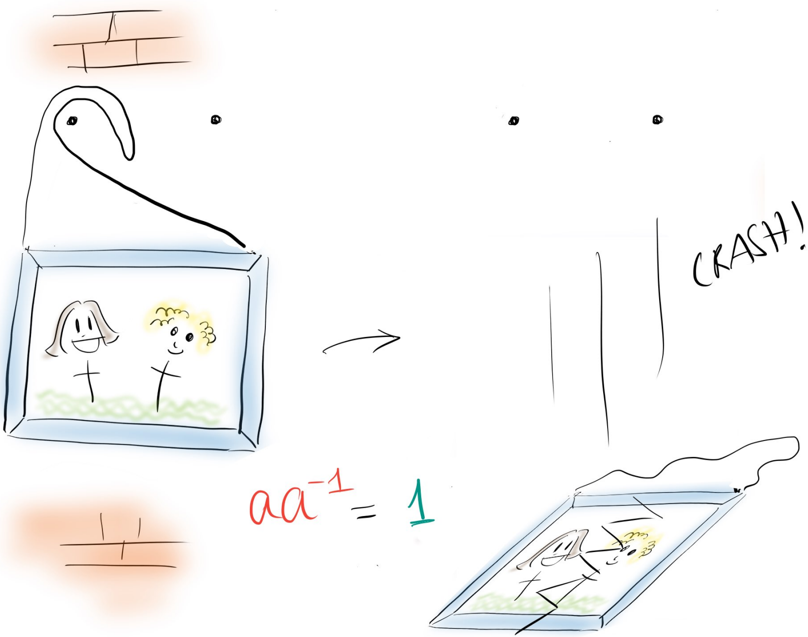

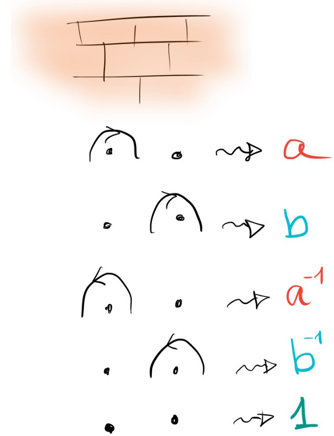

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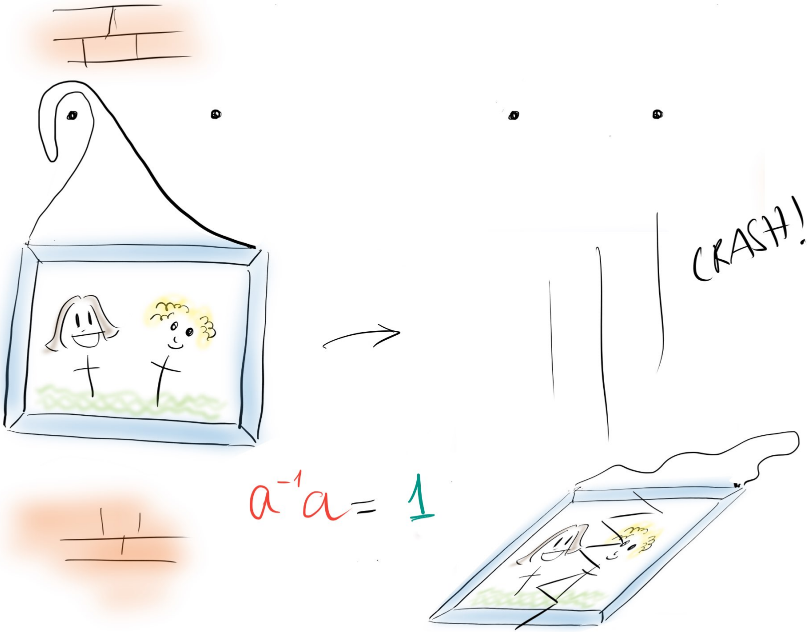
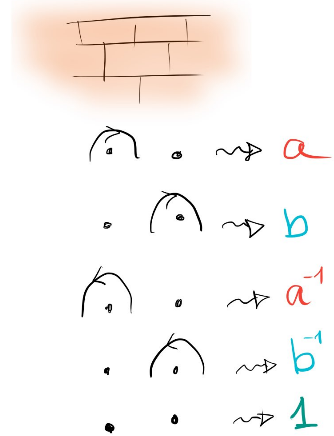


$$a^{-1}a = 1$$



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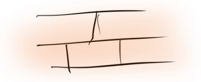
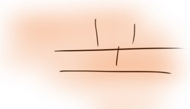
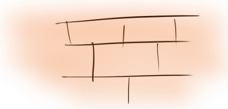
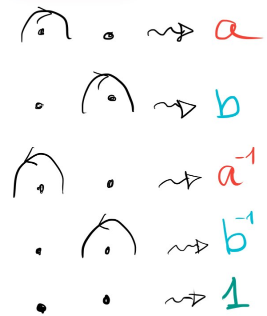




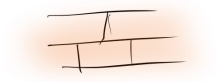


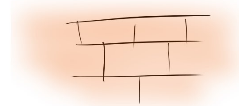


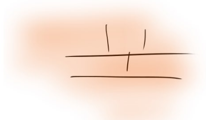
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

$\rightsquigarrow 1$

Doing nothing $\begin{cases} aa^{-1} = a^{-1}a = 1 \\ bb^{-1} = b^{-1}b = 1 \end{cases}$

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

Picture-hanging

$$abb^{-1}a = a1a = aa$$

$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

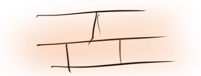
$\rightsquigarrow b^{-1}$

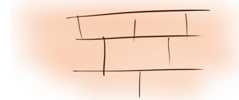
$\rightsquigarrow 1$

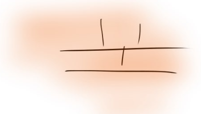
Doing nothing
$$\left\{ \begin{array}{l} aa^{-1} = a^{-1}a = 1 \\ bb^{-1} = b^{-1}b = 1 \end{array} \right.$$

Removing nail ?

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







a

b

a^{-1}

b^{-1}

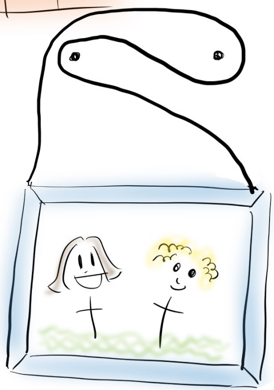
1

$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

Removing nail ?

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



aba^{-1}

$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

$\rightsquigarrow 1$

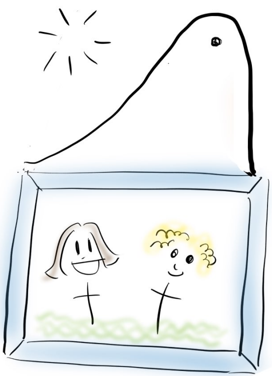
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

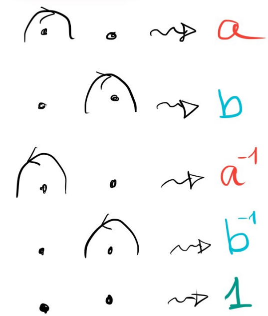
Removing nail ?

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

aba^{-1}



b



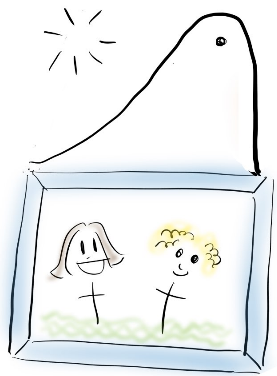
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

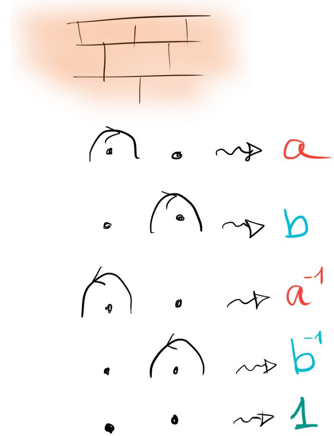
Removing nail? Erasing colour!

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

aba^{-1}



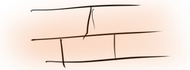
b

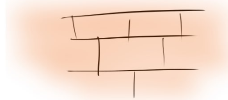


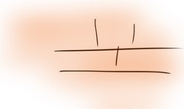
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







 ⇝ a

 ⇝ b

 ⇝ a^{-1}

 ⇝ b^{-1}

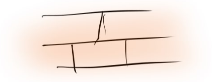
 ⇝ 1

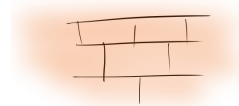
$$aa^{-1} = a^{-1}a = 1$$

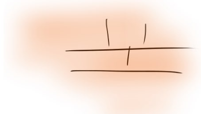
$$bb^{-1} = b^{-1}b = 1$$

same as

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

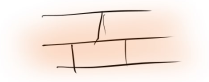
$\rightsquigarrow 1$

$$aa^{-1} = a^{-1}a = 1$$

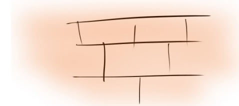
$$bb^{-1} = b^{-1}b = 1$$

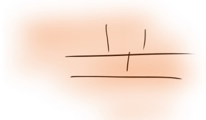
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

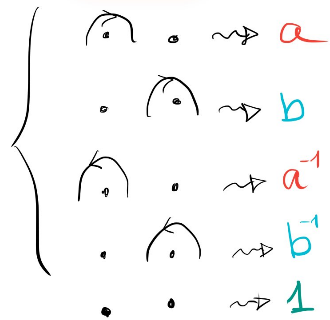
same as



- Find a word using a, a^{-1}, b, b^{-1}





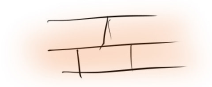


$$aa^{-1} = a^{-1}a = 1$$

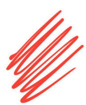
$$bb^{-1} = b^{-1}b = 1$$

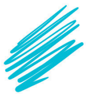
GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

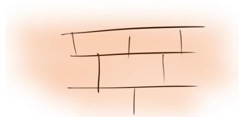
same as

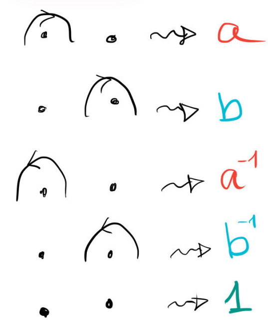


- Find a word using a, a^{-1}, b, b^{-1}
- So that if you erase one colour...



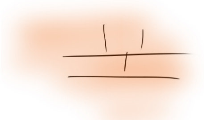






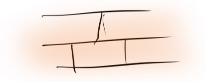
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

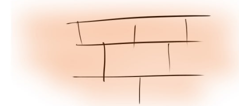


GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

same as



- Find a word using a, a^{-1}, b, b^{-1}
- So that if you erase one colour...
- and use the cancellations ...



$\rightsquigarrow a$

$\rightsquigarrow b$

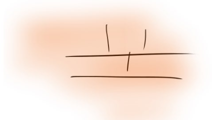
$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

$\rightsquigarrow 1$

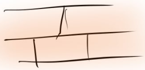
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$



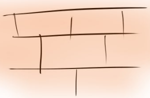
GOAL: Hang picture so that it falls if you remove *any* of the 2 nails

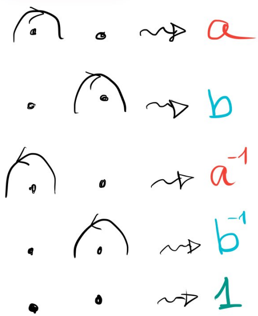
same as



- Find a word using a, a^{-1}, b, b^{-1}
- So that if you erase one colour...
- and use the cancellations ...

The word will cancel out!





$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

Sol: $aba^{-1}b^{-1}$

same as

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

- Find a word using a, a^{-1}, b, b^{-1}
- So that if you erase one colour...
- and use the cancellations ...

The word will cancel out!

$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

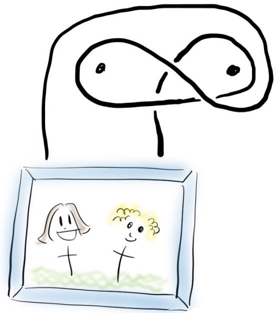
$\rightsquigarrow b^{-1}$

$\rightsquigarrow 1$

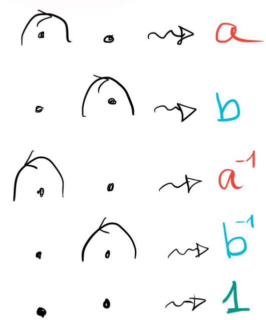
$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

Sol: $aba^{-1}b^{-1}$



GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

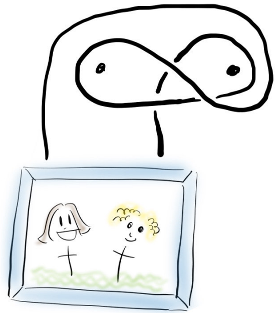


$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

Sol: $aba^{-1}b^{-1}$

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

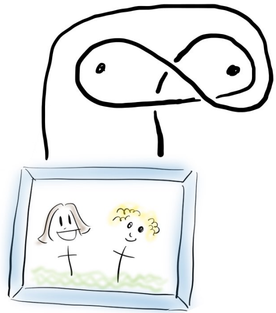
$\rightsquigarrow 1$

$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

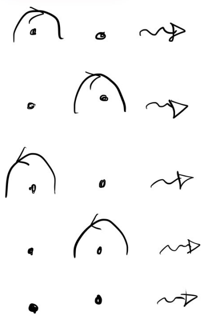
$aba^{-1}b^{-1}$

$$\boxed{\text{Sol}: aba^{-1}b^{-1}}$$



GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails

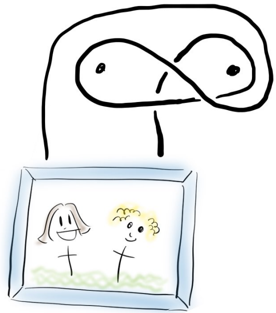
$$aba^{-1}b^{-1} \xrightarrow{\text{erase}} bb^{-1} = 1$$



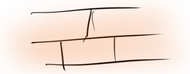
$$aa^{-1} = a^{-1}a = 1$$

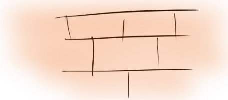
$$bb^{-1} = b^{-1}b = 1$$

$$\boxed{\text{Sol}: aba^{-1}b^{-1}}$$

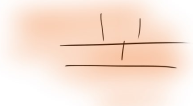


GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails





$$aba^{-1}b^{-1} \xrightarrow{\text{erase}} bb^{-1} = 1$$



$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

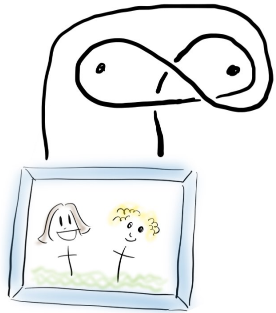
$\rightsquigarrow b^{-1}$

$\rightsquigarrow 1$

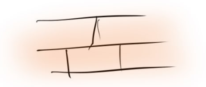
$$aa^{-1} = a^{-1}a = 1$$

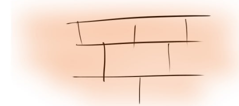
$$bb^{-1} = b^{-1}b = 1$$

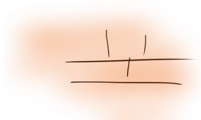
Sol: $aba^{-1}b^{-1}$



GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails







$$aba^{-1}b^{-1} \xrightarrow{\text{erase } a} bb^{-1} = 1$$

$$aba^{-1}b^{-1} \xrightarrow{\text{erase } b} aa^{-1} = 1$$

$\rightsquigarrow a$

$\rightsquigarrow b$

$\rightsquigarrow a^{-1}$

$\rightsquigarrow b^{-1}$

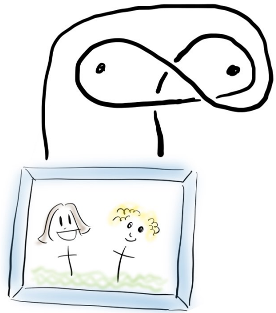
$\rightsquigarrow 1$

$aa^{-1} = a^{-1}a = 1$

$bb^{-1} = b^{-1}b = 1$

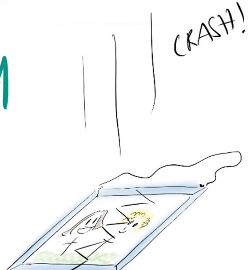
Sol: $aba^{-1}b^{-1}$

GOAL: Hang picture so that it falls if you remove <u>any</u> of the 2 nails



$$aba^{-1}b^{-1} \xrightarrow{\text{erase}} bb^{-1} = 1$$

$$aba^{-1}b^{-1} \xrightarrow{\text{erase}} aa^{-1} = 1$$



$\rightsquigarrow a$

$\rightsquigarrow b$

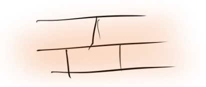
$\rightsquigarrow a^{-1}$

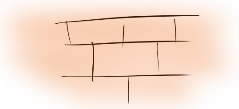
$\rightsquigarrow b^{-1}$

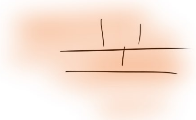
$\rightsquigarrow 1$

$$aa^{-1} = a^{-1}a = 1$$

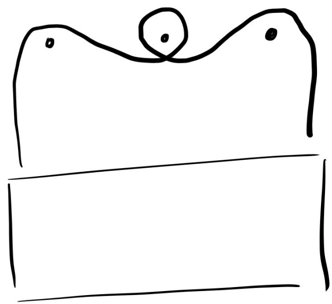
$$bb^{-1} = b^{-1}b = 1$$



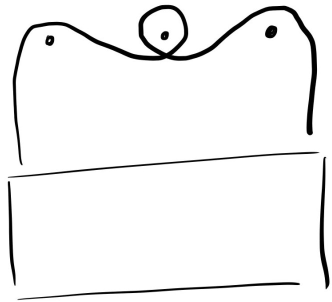




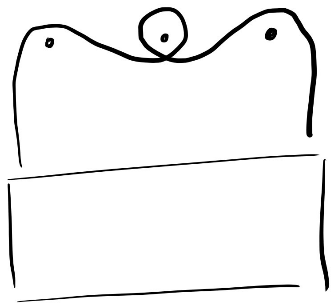
Can you do it with 3 nails?



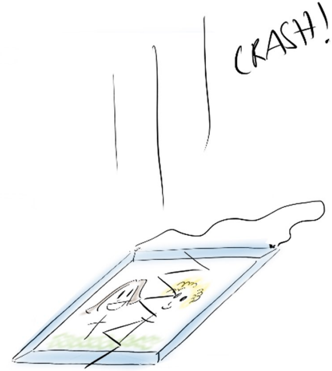
Can you do it with 5 nails?



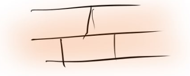
Can you do it with 5 nails?

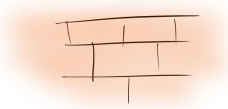


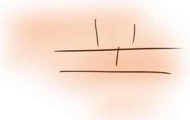
THANKS :)



(Mathematics)







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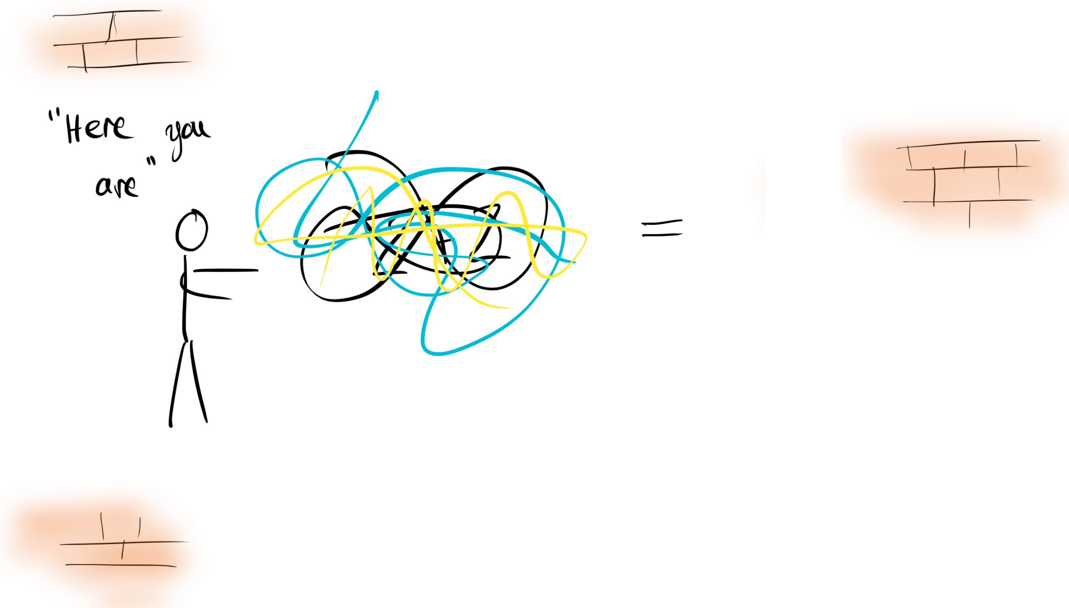

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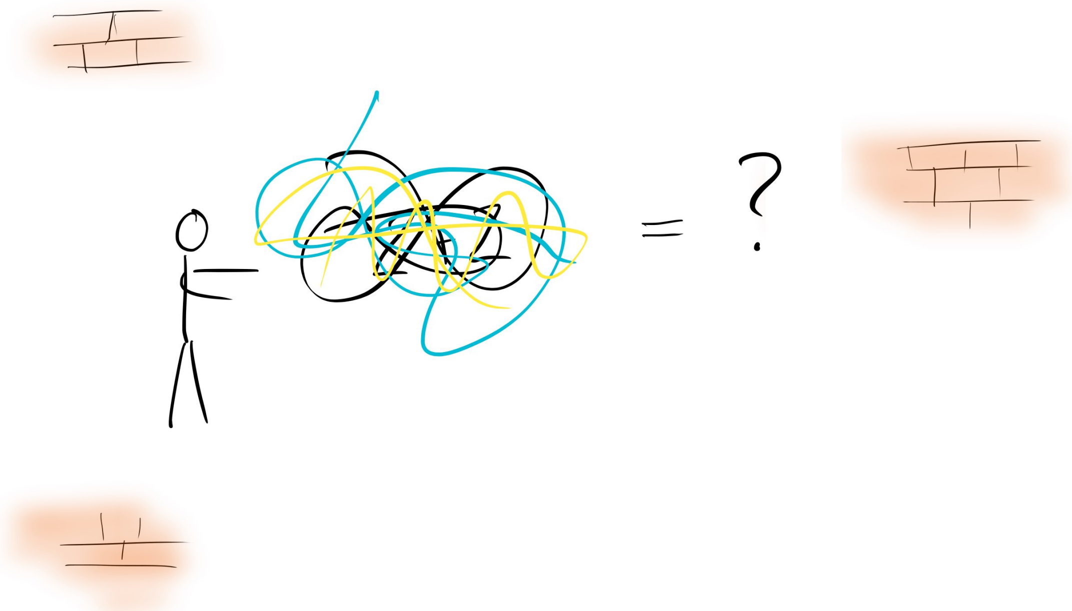

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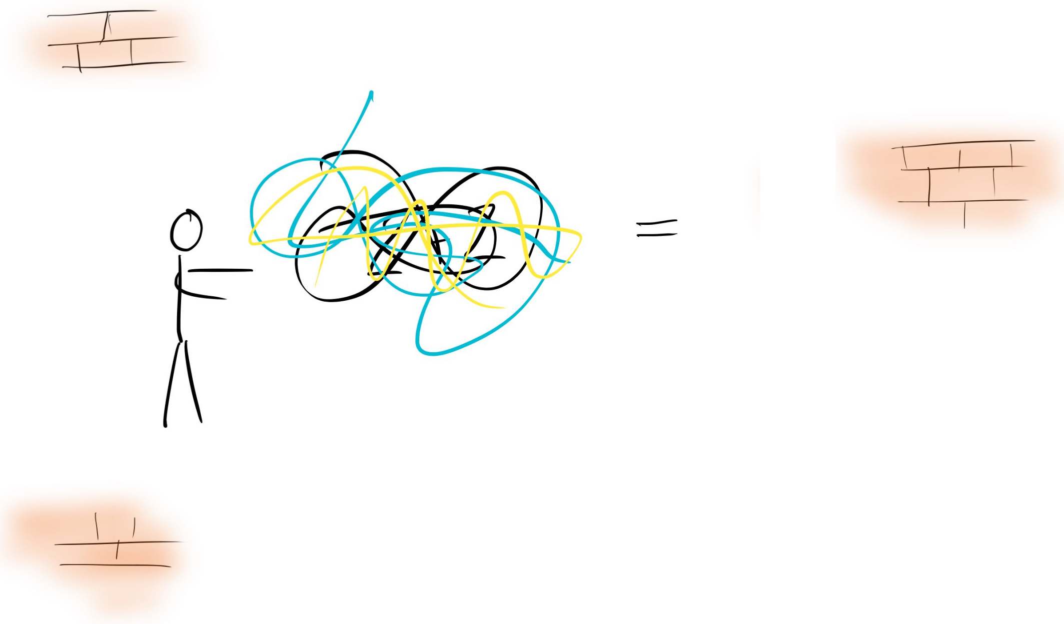

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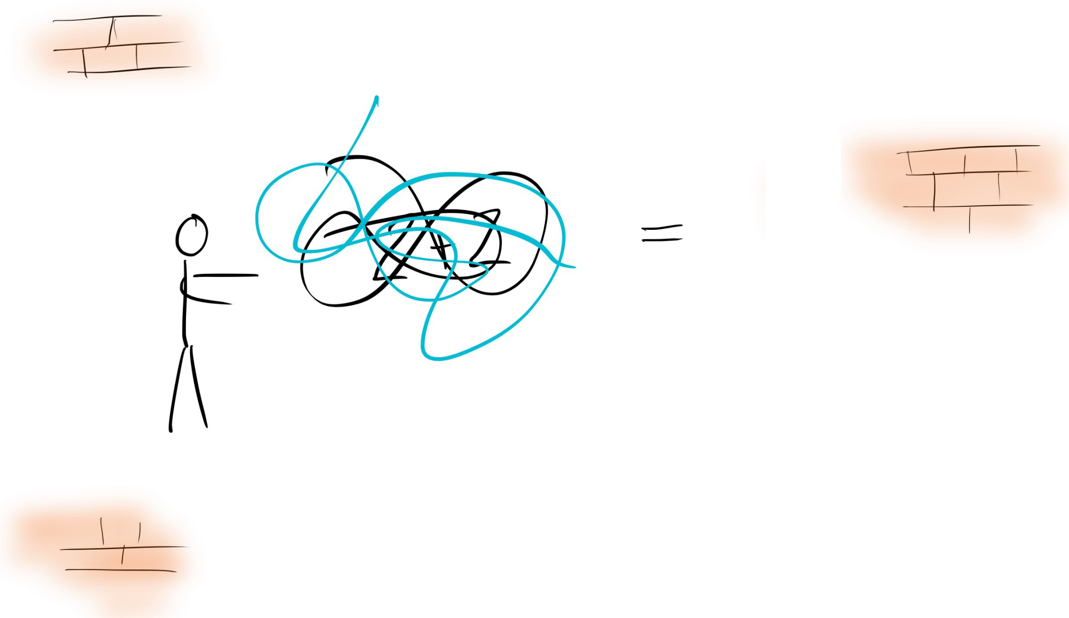

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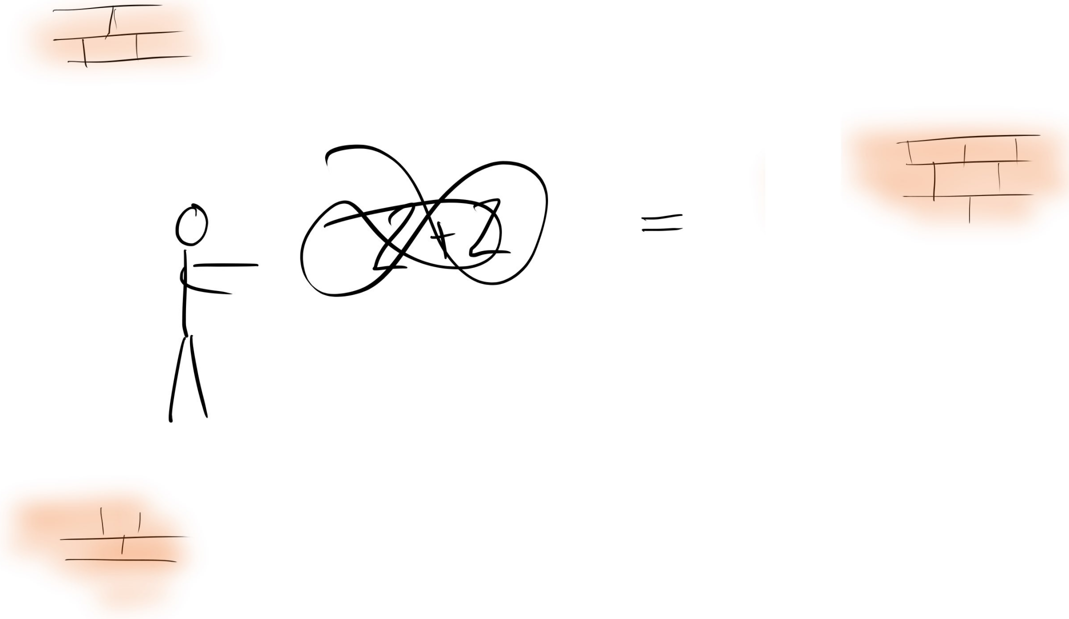
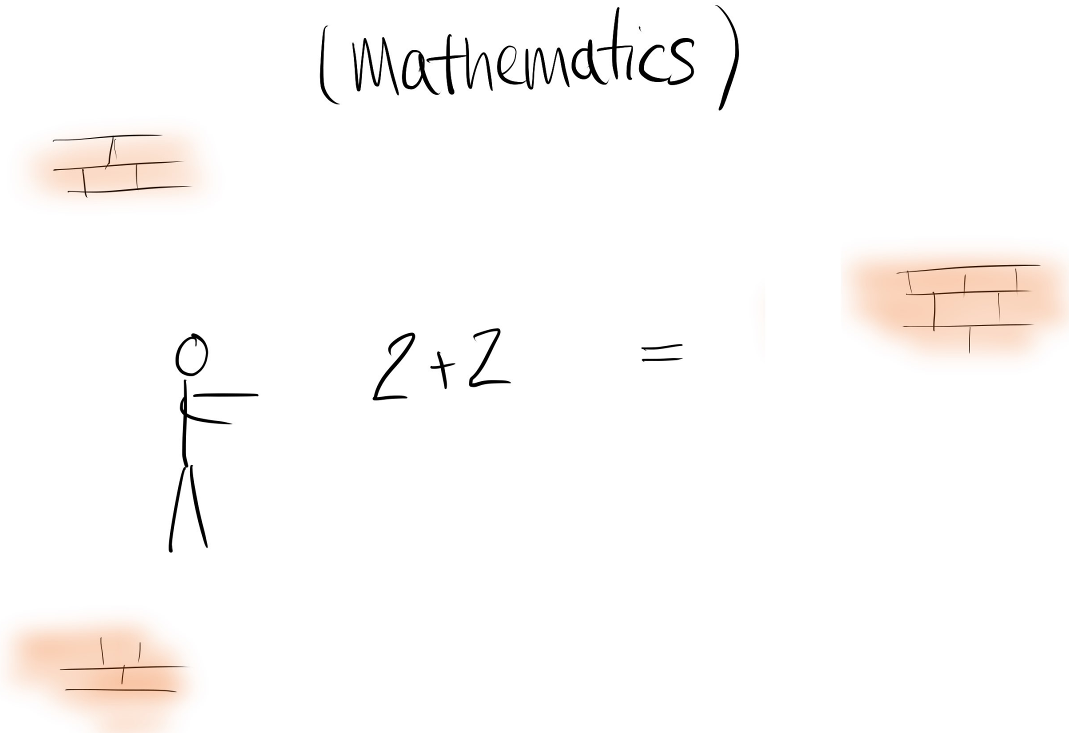
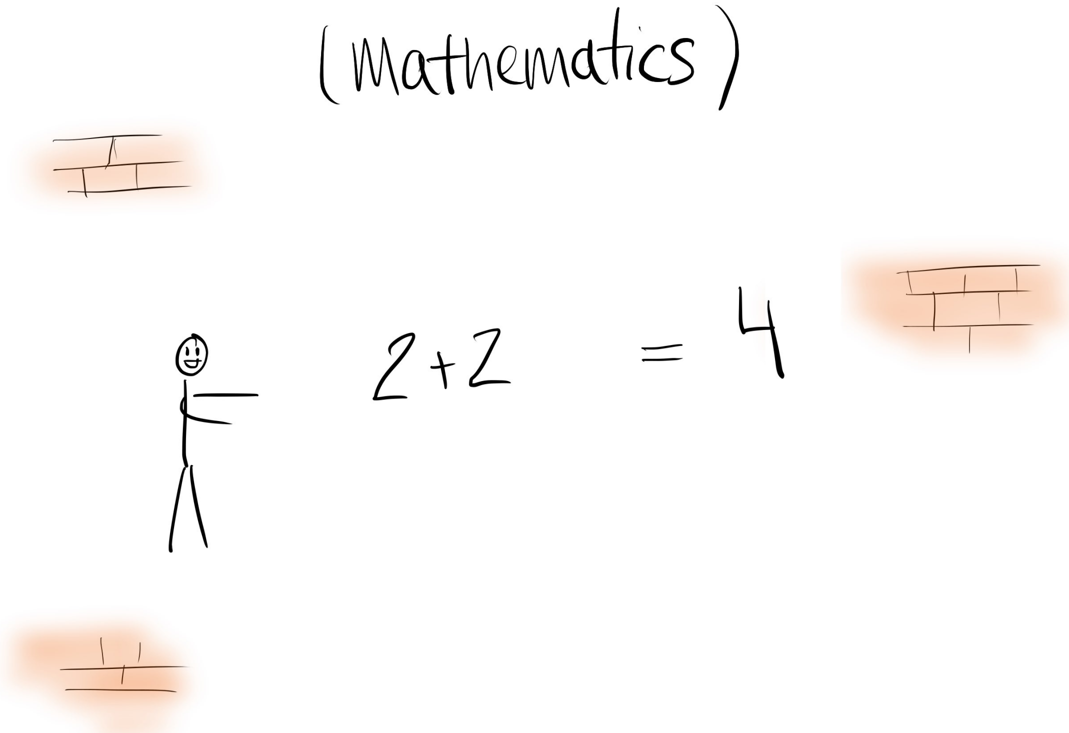



(Mathematics)
2+2 =


(Mathematics)
2+2 = 4

(Mathematics)